L12 LINUX PROJECT

Anime Episode Tracker

Bash Shell Scripting • CSV Data Handling • Cron Scheduling • Backup & Logging

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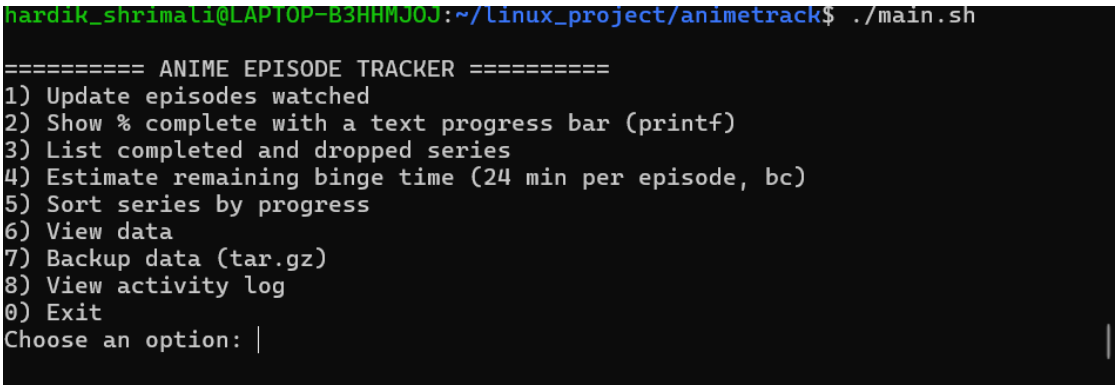
Project Overview

Anime Episode Tracker is a menu-driven Linux command-line application written in Bash. It stores anime episode data in a CSV file and provides options to update watched episodes, display progress, list completed or dropped series, estimate remaining viewing time, sort by progress, view data, create backups, and inspect activity logs.

Project directory: `~/linux_project/animetrack`

Problem statement: manually tracking episode progress across many anime series can be inconvenient. This project centralizes episode counts and status in a CSV file and provides quick progress calculations and simple automation.

Project screenshot — main menu



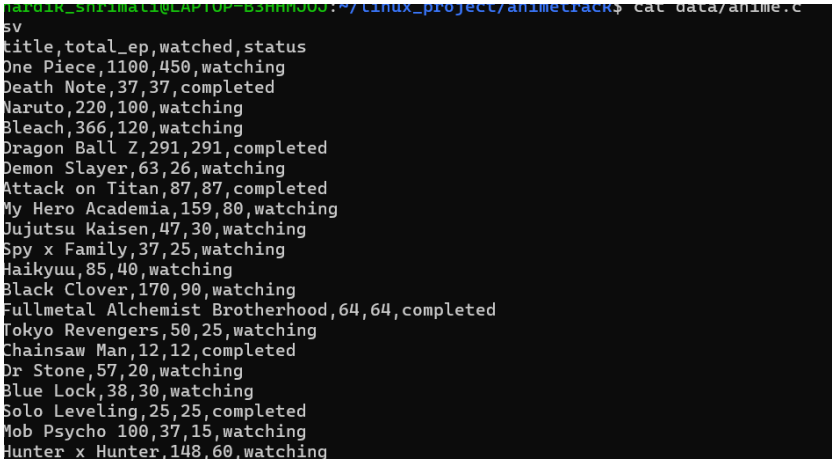
1. Project Structure and Data

The project separates its executable script, reusable Bash functions, CSV data, reports, backups, logs, and README documentation.

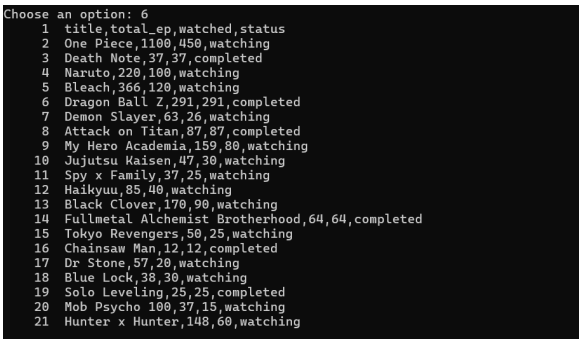
```
animetrack/
■■■ main.sh
■■■ lib/functions.sh
■■■ data/anime.csv
■■■ reports/
■■■ backups/
■■■ logs/
■■■ README.md
```

CSV schema: `title,total_ep,watched,status`. The displayed dataset contains 20 anime records plus the header row.

Screenshot — CSV data displayed in terminal



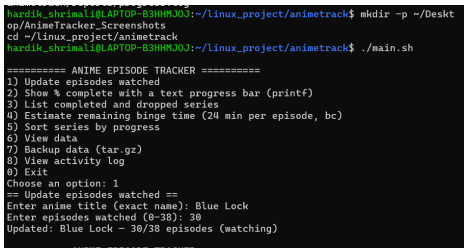
Screenshot — application data view



2. Core Features (1–3)

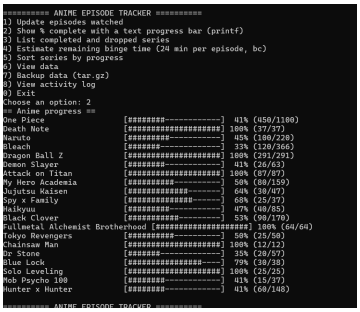
Feature 1 — Update Episodes Watched

The user enters an exact anime title and the watched episode count. The screenshot shows Blue Lock updated to 30 of 38 episodes.



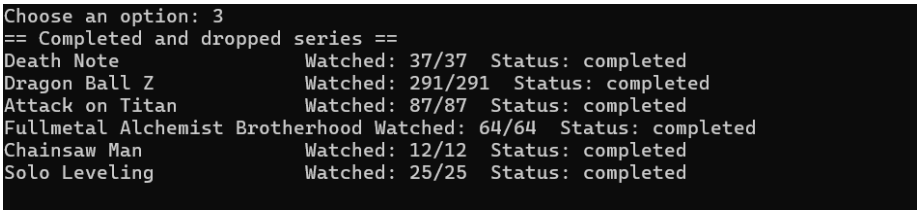
Feature 2 — Progress Percentage and Text Bar

Displays the watched/total count, percentage completion, and a text-based progress bar for each series.



Feature 3 — Completed and Dropped Series

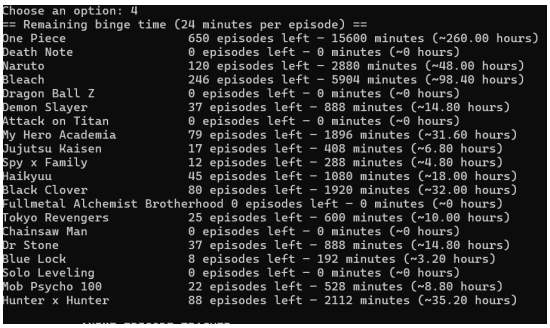
Lists series marked completed or dropped. The captured output demonstrates completed series and their watched counts.



3. Core Features (4–6)

Feature 4 — Remaining Binge Time

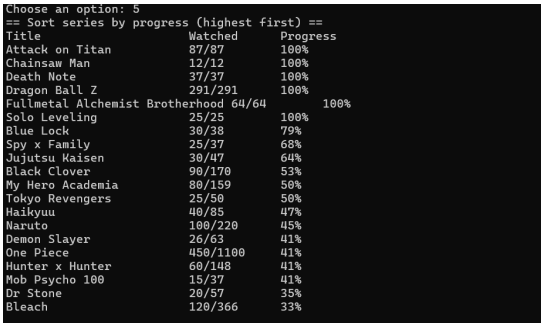
Estimates unwatched episodes and remaining viewing time using 24 minutes per episode.



```
Choose an option: 4
== Remaining binge time (24 minutes per episode) ==
One Piece                   650 episodes left — 15600 minutes (~260.00 hours)
Death Note                  0 episodes left — 0 minutes (~0 hours)
Naruto                      120 episodes left — 2880 minutes (~48.00 hours)
Bleach                      246 episodes left — 5904 minutes (~98.40 hours)
Dragon Ball Z               0 episodes left — 0 minutes (~0 hours)
Demon Slayer                37 episodes left — 888 minutes (~14.80 hours)
Attack on Titan             0 episodes left — 0 minutes (~0 hours)
My Hero Academia            79 episodes left — 1896 minutes (~31.60 hours)
Jujutsu Kaisen              17 episodes left — 408 minutes (~6.80 hours)
Spy x Family                12 episodes left — 288 minutes (~4.80 hours)
Haikyuu                     45 episodes left — 1080 minutes (~18.00 hours)
Black Clover                80 episodes left — 1920 minutes (~32.00 hours)
Fullmetal Alchemist Brotherhood 0 episodes left — 0 minutes (~0 hours)
Tokyo Revengers             25 episodes left — 600 minutes (~10.00 hours)
Chainsaw Man                0 episodes left — 0 minutes (~0 hours)
Dr Stone                    37 episodes left — 888 minutes (~14.80 hours)
Blue Lock                   8 episodes left — 192 minutes (~3.20 hours)
Solo Leveling               0 episodes left — 0 minutes (~0 hours)
Mob Psycho 100              22 episodes left — 528 minutes (~8.80 hours)
Hunter x Hunter             88 episodes left — 2112 minutes (~35.20 hours)
```

Feature 5 — Sort by Progress

Sorts anime series by progress percentage, with the highest-progress series first.



```
Choose an option: 5
== Sort series by progress (highest first) ==
Title                       Watched       Progress
Attack on Titan             87/87         100%
Chainsaw Man                12/12         100%
Death Note                  37/37         100%
Dragon Ball Z               291/291       100%
Fullmetal Alchemist Brotherhood 64/64         100%
Solo Leveling               25/25         100%
Blue Lock                   30/38         79%
Spy x Family                25/37         68%
Jujutsu Kaisen              30/47         64%
Black Clover                90/170        53%
My Hero Academia            80/159        50%
Tokyo Revengers             25/50         50%
Haikyuu                     40/85         47%
Naruto                      100/220       45%
Demon Slayer                26/63         41%
One Piece                   450/1100      41%
Hunter x Hunter             60/148        41%
Mob Psycho 100              15/37         41%
Dr Stone                    20/57         35%
Bleach                      120/366       33%
```

Feature 6 — View Data

Displays the CSV data from the terminal for a quick view of the stored records.

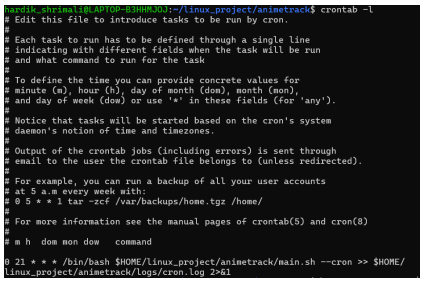
4. Cron Scheduling, Logs and Submission

Cron Scheduling

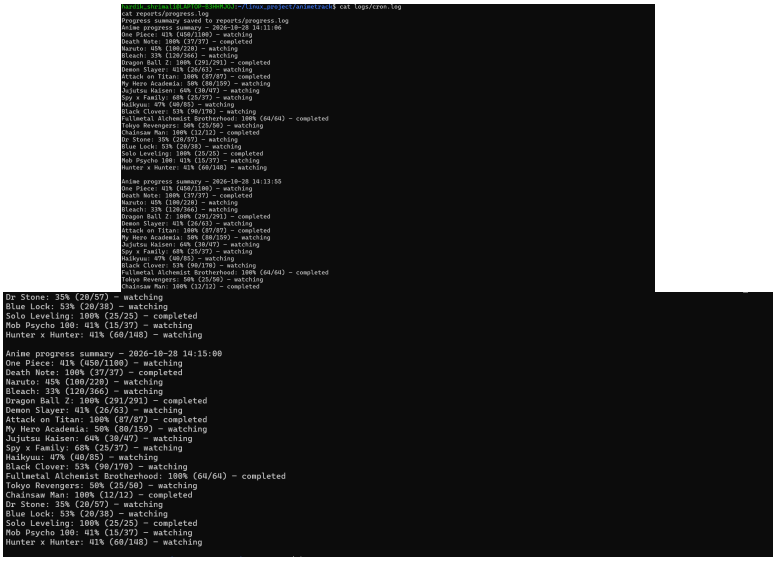
The application supports a non-interactive cron mode using `main.sh --cron`. The configured crontab entry runs daily at 21:00 system time and redirects command output to logs/cron.log.

```
0 21 * * * /bin/bash $HOME/linux_project/animetrack/main.sh --cron >>
$HOME/linux_project/animetrack/logs/cron.log 2>&1
```

Screenshot — crontab -l



Cron output and generated progress summary



Learning Outcomes

- Practiced Bash scripting, functions, loops, conditions, and input validation.
- Practiced reading and updating CSV data with shell commands.
- Calculated progress percentages and remaining viewing time.
- Used cron scheduling, activity logging, and tar.gz archives.

Submission checklist: include main.sh, lib/functions.sh, data/anime.csv, reports/, logs/, README.md, and this report in the final project archive. The terminal screenshot shows an archive listing; rebuild the archive after placing this final PDF in reports/ so the PDF is included.